The halting function h is not computable

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-computability-computability-theory/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the definition-by-cases and the primitive recursive g are both explicit in-span; no step is delegated

Editorial note. The label thm:halting-problem names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- No universal computable function. Same unit, \ollabel{thm:no-univ} (source lines 1612-1618). The contradiction target: if h were computable, Un' would be a universal computable function for the total partial computable functions, contradicting the stub.

Source reference. Original theorem thm-no-univ: sources/olp-computability-computability-theory/pinned-original-source.tex, source lines 1612–1618. Prerequisite pointer invoked in the counted proof; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Let*

$$h(e,x) = \begin{cases} 1 & \textit{if } \mathrm{Un}(e,x) \textit{ is defined} \\ 0 & \textit{otherwise.} \end{cases}$$

Then h is not computable.

Proof. Suppose h is computable. We show that this would let us define a universal computable function. Define

$$\mathrm{Un'}(e,x) = \begin{cases} \mathrm{Un}(e,x) & \text{if } h(e,x) = 1 \\ 0 & \text{otherwise.} \end{cases}$$

But now Un'(e, x) is a total function, and is computable if h is. For instance, we could define g using primitive recursion, by

$$g(0, e, x) \simeq 0$$
$$g(y + 1, e, x) \simeq \mathrm{Un}(e, x);$$

then

$$\mathrm{Un}'(e, x) \simeq g(h(e, x), e, x).$$

Since Un'(e, x) agrees with $\mathrm{Un}(e, x)$ wherever the latter is defined, Un' is universal for those partial computable functions that happen to be total. But this contradicts thm-no-univ. □